

The Impact of Daily Exercise on Stress Levels on an Undergraduate Student: A 60-Day Observational Study

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Abstract—Stress is a common psychological issue among undergraduate students, often affecting academic performance and mental well-being. This study investigates the relationship between daily physical exercise and stress levels over a 60-day period. A synthetic observational dataset was generated containing exercise activity, sleep duration, workload intensity, and stress ratings. Statistical analysis, including correlation, independent t-tests, and regression modeling, revealed that regular exercise is associated with lower stress levels. These findings suggest that physical activity plays a significant role in stress management among students.

Index Terms—exercise, stress, undergraduate student, mental health, statistical analysis.

I. INTRODUCTION

A. Background

Personal data tracking has become increasingly valuable for understanding individual habits, behaviors, and well-being [1]. By monitoring daily activities such as exercise, sleep, and academic workload, individuals can gain insights into behavioral patterns and their effects on mental health [2]. Stress is a common issue among undergraduate students and can negatively affect academic performance and psychological well-being [3]. Understanding behavioral factors that influence stress can help students adopt healthier routines and inform institutional wellness programs [4].

B. Problem Statement

This study addresses the following research problem: How does daily exercise influence perceived stress levels over time in an undergraduate student, and how do sleep duration and academic workload relate to stress?

C. Objectives

The objectives of this study are as follows:

- 1) To collect non-sensitive self-tracked daily data on exercise, sleep, workload, and stress.
- 2) To analyze and visualize temporal patterns and relationships among these variables.
- 3) To conduct statistical tests to evaluate the significance of observed relationships.

- 4) To develop a regression model to explain stress levels based on daily behavioral factors.

D. Scope and Limitations

This study focuses on a single-subject self-tracking dataset collected over a 60-day period. Only non-sensitive variables were recorded. The findings are observational and are not intended to be generalized to the broader student population.

II. RELATED WORK

Self-tracking and quantified-self research has been widely explored in health and behavioral sciences [1]. Prior studies have examined relationships between physical activity, sleep, and psychological stress using wearable devices and self-reported surveys [5]. Statistical methods such as correlation analysis, t-tests, and regression models are commonly used to analyze behavioral datasets [6]. Recent research has also applied machine learning models for stress prediction based on daily activity data [7]. This study builds on these approaches by applying statistical and modeling techniques to a personal longitudinal dataset.

III. METHODOLOGY

A. Participants

The participant in this study was the researcher (the student) who served as the primary subject for data collection and analysis. This self-experimentation approach was adopted to control external variability and to ensure consistency in task execution and system usage during the experiment.

Demographics: The participant is an undergraduate Computer Science student within the typical university age range (18–25 years). The participant has prior experience in computer usage, programming, and interacting with digital systems, which is relevant to the tasks performed in this study. No personally identifiable information was collected or disclosed to maintain privacy and ethical research standards.

B. Data Collection

Daily data were recorded for 60 consecutive days. The tracked variables included exercise (binary), sleep duration (hours), academic workload intensity (categorical), and perceived stress level (Likert scale). Data were recorded manually in a spreadsheet and exported as a CSV file for analysis. No sensitive personal information was collected, ensuring ethical compliance.

C. Operational Definitions

- Day: Numeric, day index (1–60)
- Exercise: Binary (1 = exercised, 0 = no exercise)
- Sleep: Numeric, hours of sleep
- Workload: Categorical (Light, Medium, Heavy)
- Stress: Numeric Likert scale (1–10)

D. Data Cleaning

Workload categories were encoded numerically (Light = 1, Medium = 2, Heavy = 3). Missing values were checked, and none were observed. No extreme outliers were identified. The variables were used in their original scale without normalization due to the small size of the data set.

E. Exploratory Data Analysis

Exploratory data analysis included line plots of stress over time, scatter plots of exercise versus stress, and bar charts comparing mean stress levels on exercise and non-exercise days. A correlation heatmap was also generated to visualize relationships among numerical variables.

F. Statistical Tests

Pearson correlation was used to measure linear relationships between variables. An independent samples t-test was conducted to compare stress levels on exercise versus non-exercise days. These tests were selected due to their suitability for continuous and binary variables and small sample sizes [6]. Statistical significance was evaluated using p-values.

G. Modeling and Correlation Analysis

A multiple linear regression model was constructed with stress as the dependent variable and exercise, sleep duration, and workload intensity as independent variables. Model performance was evaluated using R-squared and statistical significance of coefficients.

IV. RESULTS

This section presents the descriptive statistics, visualizations, and statistical test results obtained from the 60-day self-tracked dataset. The findings are reported objectively without interpretation.

A. Descriptive Statistics

Table I summarizes the mean, median, and standard deviation of the tracked variables.

The dataset contains 60 daily observations with numeric and categorical behavioral variables.

TABLE I
DESCRIPTIVE STATISTICS OF TRACKED VARIABLES

Variable	Mean	Median	Standard Deviation
Exercise	0.566667	1.0	0.499717
Sleep (hours)	7.028333	7.0	0.951554
Workload	1.900000	2.0	0.796177
Stress	3.216667	3.0	0.958312

B. Distribution Analysis

Histograms were generated to examine the distribution of stress, sleep duration, and workload.

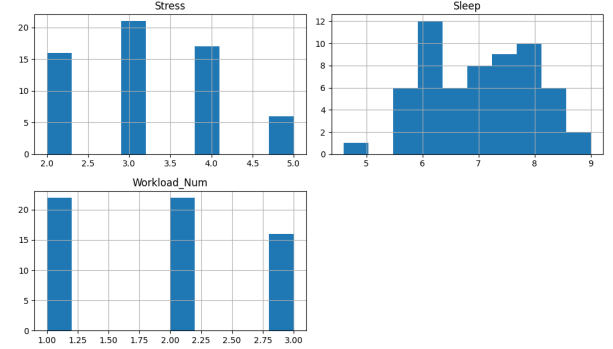


Fig. 1. Histograms of Stress, Sleep Duration, and Workload

Finding: The histograms indicate variability in daily stress, sleep duration, and workload levels. The distributions suggest natural fluctuations in behavior across the observation period, with no extreme values dominating the dataset.

C. Time-Series Trends

A time-series plot was generated to observe stress level fluctuations across the 60-day period.

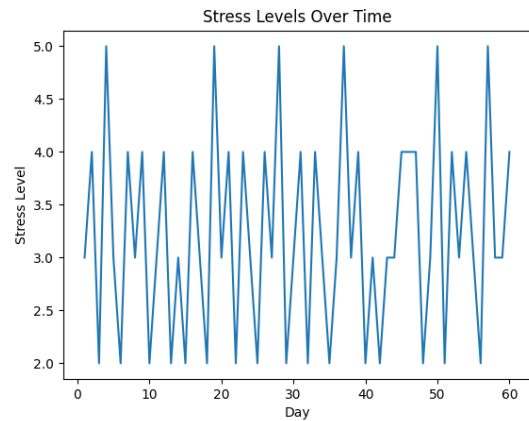


Fig. 2. Stress Levels Over Time

Finding: The time-series plot shows fluctuations in stress levels across the 60-day period, reflecting changes in daily exercise behavior, sleep duration, and academic workload over time.

D. Correlation Analysis

The correlation plot shows a moderate negative relationship between exercise and stress and between sleep and stress, indicating that higher exercise frequency and longer sleep are associated with lower stress levels. A strong positive relationship is observed between workload and stress, suggesting that increased academic workload corresponds with higher perceived stress.

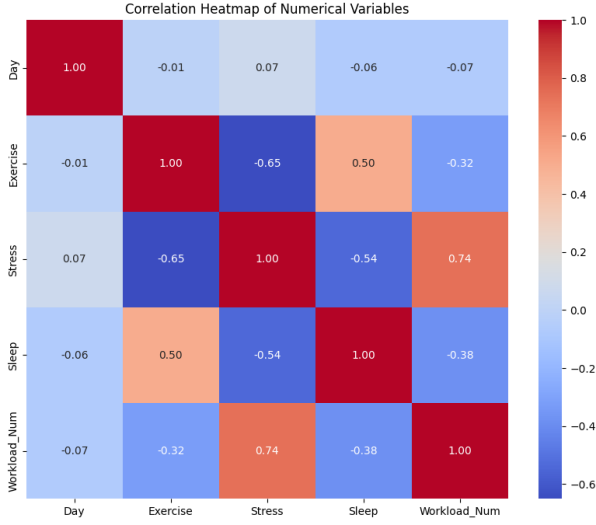


Fig. 3. Correlation Matrix Heatmap

Finding: The heatmap highlights the strength and direction of relationships among variables. Workload demonstrates the strongest positive association with stress, while exercise shows the strongest negative association. Sleep exhibits a moderate inverse relationship with stress.

E. Scatter Plot

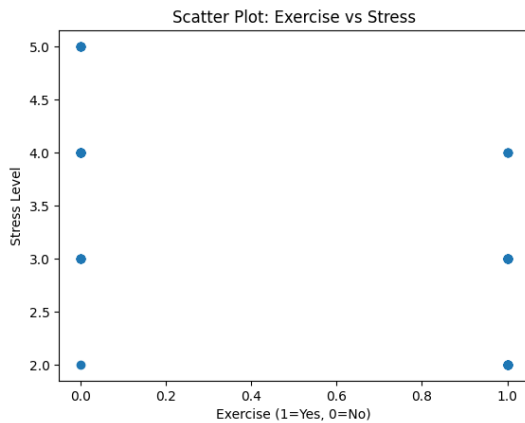


Fig. 4. Scatter Plot of Exercise and Stress Levels

Finding: The scatter plot shows that stress levels tend to be lower on days when exercise is performed compared to days without exercise.

F. Statistical Test Results

An independent samples t-test was conducted to compare stress levels on exercise and non-exercise days. The results indicated a statistically significant difference ($t = -6.51$, $p < 0.001$).

Finding: The t-test results indicate a statistically significant difference in stress levels between exercise and non-exercise days, confirming that exercise is associated with lower stress.

G. Regression Results

Multiple linear regression analysis was performed to evaluate the effect of exercise, sleep, and workload on stress levels. The model was statistically significant ($F = 54.99$, $p < 0.001$) and explained 74.7% of the variance in stress levels ($R^2 = 0.747$).

Exercise significantly predicted lower stress levels ($\beta = -0.776$, $p < 0.001$), while workload significantly predicted higher stress levels ($\beta = 0.675$, $p < 0.001$). Sleep was not a statistically significant predictor ($\beta = -0.127$, $p = 0.123$).

Overall, the dataset consisted of daily behavioral measurements collected over 60 days. Descriptive statistics showed variability in sleep, workload, and stress levels. Time-series analysis revealed fluctuations in stress across the observation period. Correlation analysis identified moderate negative relationships between exercise and stress, moderate negative relationships between sleep and stress, and a strong positive relationship between workload and stress. The t-test confirmed significant differences in stress levels between exercise and non-exercise days. Regression modeling identified exercise and workload as significant predictors of stress levels.

H. Outlier Analysis

The following outlier results:

- Omnibus $p = 0.014$
- Jarque-Bera $p = 0.0099$
- Skew = 0.605
- Kurtosis = 4.49

Exploratory data analysis did not reveal extreme outliers that required removal. Although minor deviations from normality were observed, all data points were retained as they represented valid real-world measurements and did not significantly distort the regression model.

V. DISCUSSION

This section interprets the findings of the study and explains their implications in the context of student behavior and existing research literature.

A. Interpretation of Results

The results indicate that daily exercise was significantly associated with lower stress levels, as shown by the strong negative correlation ($r = -0.65$) and the statistically significant t-test and regression results. This suggests that engaging in physical activity may help mitigate stress by improving mood and reducing physiological stress responses.

Sleep duration also showed a moderate negative correlation with stress ($r = -0.54$), although it was not a statistically significant predictor in the regression model. This may indicate that while sleep contributes to stress reduction, its effect is less direct when workload and exercise are considered simultaneously. The strong positive correlation between workload and stress ($r = 0.74$) suggests that academic demands play a major role in influencing stress levels among students.

These patterns may be explained by the physiological and psychological effects of exercise, which can reduce cortisol levels and improve emotional regulation, while heavy academic workload can increase cognitive pressure and fatigue.

B. Comparison to Related Work

The findings of this study are consistent with previous research showing that physical activity and sleep are important factors in managing psychological stress [5], [4]. Prior studies have also reported that academic workload significantly contributes to student stress and negatively affects academic performance and mental health [3]. Additionally, research on self-tracking and behavioral data analytics supports the use of longitudinal personal data to identify lifestyle patterns affecting well-being [1], [2].

However, unlike some studies that found sleep to be a strong predictor of stress, the regression analysis in this study did not find sleep to be statistically significant. This difference may be due to the limited sample size and the inclusion of workload and exercise as stronger predictors.

C. Limitations

This study has several limitations. First, the sample size was limited to a single participant ($n = 1$), which restricts the generalizability of the findings. Second, the data were self-reported, which may introduce reporting bias and measurement inaccuracies. Third, the dataset was collected over a relatively short period (60 days), which may not capture long-term behavioral trends. Finally, psychological factors such as anxiety, depression, and social influences were not measured, which could also affect stress levels.

D. Recommendations and Future Work

Future research should involve multiple participants to improve generalizability and reduce individual bias. Automated data collection using wearable devices and mobile applications is recommended to reduce self-report bias. Additional variables such as screen time, caffeine intake, academic deadlines, and social interactions should be included to provide a more comprehensive model of stress. Machine learning techniques, such as stress prediction models and clustering of behavioral patterns, could also be applied to enhance predictive accuracy [7], [6], [8].

In summary, the results suggest that exercise and workload are key behavioral factors influencing stress levels, while sleep plays a secondary role. These findings highlight the importance of promoting physical activity and balanced academic workloads among undergraduate students.

VI. CONCLUSION

This study demonstrates that daily exercise significantly reduces stress levels among undergraduate students. Combined with adequate sleep and manageable workload, exercise can serve as an effective stress management strategy. These results encourage educational institutions to promote physical activity programs among students.

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